When Nothing Happened: Negation, Time Frames, and Event Anaphora in English and Mandarin [1]

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[1] I am grateful to ...

Judgments and Discussions: Yi Wu (Eastern Chinese), Xuannan Yang (Southern Chinese), Daniel Worthing (British English), NUS Syn-Sem Reading Group, UBC Topics in Semantics course (Winter 2026), ICFL-11...

1 Introduction

1.1 Background and English Data

Negation blocks discourse referents inside its scope from anaphoric accessibility. This accounts for the infelicity of discourses such as # *John did not buy a donkey. It is black* – the indefinite in the negation does not introduce an accessible discourse referent, the pronoun *it* fails to locate a φ-matching antecedent.

Anaphora is not limited to nominal reference. It is well established that event referents also enter into anaphoric relations that underlie temporal sequencing: in narratives, each event is typically understood to occur 'just after'/'immediately after' the preceding one (Partee, 1973, 1984; Hinrichs, 1986; Dowty, 1986; among others).

This paper focuses on two interesting properties of event referents in narrative: (I) their anaphoric accessibility is **insensitive** to negation, and (II) it is **sensitive** to the size of time frames. Examples (1) and (2) illustrate these properties in English, respectively:

1. [1] Henry told the thief to leave. [2] The thief **did not** leave. [3] Henry shot him.
   $[\iota\, thief\ [\lambda x\ [\neg\ [\exists e.leave(x, e)]]]]$
2. a. Johnny did not see his father at noon. ✓$_J$ He then cried.
   b. Johnny did not see his father in June. ?$_J$ He then cried.[2]

Negation in (1.[2]) blocks the leaving event inside its scope. Nevertheless, [1]–[2] and [2]–[3] each establish a temporal succession: the thief's not leaving is understood to follow Henry's request, and the shooting is understood to follow the thief's not leaving. Native speakers of English report that (2a) supports a reading in which Johnny's crying immediately follows the not-seeing event. By contrast, (2b) is most naturally understood as describing a negative state spanning all of June, and the 'just-after' reading of *then* is unavailable.

The 'just-after' reading in (2b) becomes more available, however, when the context narrows down the time frame:

> [**Context**: Johnny's father is in the military and rarely comes home. He promised Johnny that he would come home once a month, on the morning of the first Sunday. In June, however, he did not come.]
>
> In June, Johnny did not see his father. ✓$_J$ He then cried.

[2] ✓$_J$ = 'just-after' reading after the negated sentence available;
?$_J$ = 'just-after' reading after the negated sentence unnatural or requires additional context.

1.2 Mandarin Data with Two Negators

1.2.1 Narratives without Time Frames

Mandarin Chinese shows similar properties, but with a contrast between its two negators, *mei(-you)* and *bu*.[3] First, consider the narrative in (3):

3. [1] Laoshi rang Zhangsan zhan qilai. [2] Ta {✓$_J$ ***mei(-you)*** / ✓$_J$ ***bu***} zhan.
   teacher ask Zhangsan stand up. 3SG {✓$_J$NEG.m / ✓$_J$NEG.b} stand.

   [3] Laoshi piping-le ta.
   teacher criticize-PFV 3SG.

   'The teacher asked Zhangsan to stand up. He did not stand up. The teacher criticized him.'

In (3), either negator can appear in [2] without disrupting the narrative, paralleling the English example in (1). The teacher's criticism in [3] is understood to immediately follow Zhangsan's not standing up in [2], regardless of which negator is used.[4]

1.2.2 Narratives With Different Time Frames

The two negators diverge, however, when the size of the time frame (underlined) varies. *Mei(-you)* in (4) patterns like English *not* in (2). With a small time frame such as *wanshang-qidian* ('7pm') in (4a), the 'just-after' reading is available: Mom's criticism is understood to immediately follow Xiaoming's not showering. With a larger time frame such as *shangge-xingqi* ('last week') in (4b), this reading is unnatural, and mom's criticism is understood to occur some time later.[5]

4. a. Xiaoming wanshang-qidian ***mei(-you)*** xizao. ✓$_J$ Mama piping-le ta.
   Xiaoming night-7.o'clock NEG.m shower. ✓$_J$ Mom criticize-PFV 3SG.

   'Xiaoming didn't shower at 7pm. (His) mom then criticized him.'

[3] Throughout this paper, NEG.m and NEG.b gloss *mei(-you)* and *bu*, respectively.

[4] Mandarin *bu* is commonly assumed to be ambiguous between volitional and habitual interpretations. In narrative contexts such as (3) and (5), however, *bu* is interpreted with the so-called 'volitional reading'; a habitual reading is difficult to obtain and makes the narrative unnatural.

[5] As in English, context can make the 'just-after' reading available for *mei(-you)* with a large time frame. One of my consultants noted that, given enough context, *mei(-you)* can also support this reading with a time frame such as *shangge-xingqi* ('last week'):

[**Context**: In winter, Xiaoming only showers once a month, always on the 15th. Last month, however, he did not shower.]

Xiaoming shangge-yue ***mei(-you)*** xizao. ✓$_J$ Mama piping-le ta.
Xiaoming up.CL-month NEG.m shower. ✓$_J$ Mom criticize-PFV 3SG.

'Xiaoming didn't shower last month. (His) mom criticized him.'

b. ✓$_J$ Xiaoming wanshang-qidian ***mei(-you)*** xizao bei Mama piping-le.
✓$_J$ Xiaoming night-7.o'clock NEG.m shower PASS Mom criticize-PFV.

'Xiaoming didn't shower at 7pm and was criticized by (his) mom.'

c. Xiaoming shangge-yue ***mei(-you)*** xizao. ?$_J$ Mama piping-le ta.
Xiaoming up.CL-month NEG.m shower. ?$_J$ Mom criticize-PFV 3SG.

'Xiaoming didn't shower last month. ?$_J$(His) mom then criticized him.'

d. ?$_J$ Xiaoming shangge-yue ***mei(-you)*** xizao bei Mama piping-le.
?$_J$ Xiaoming up.CL-month NEG.m shower PASS Mom criticize-PFV.

'Xiaoming didn't shower last month and was criticized ny (his) mom.'

Bu, by contrast, supports a 'just-after' reading with either large or small time frames:

5. a. Xiaoming wanshang-qidian ***bu*** xizao. ✓$_J$Mama piping-le ta.
Xiaoming night-7.o'clock NEG.b shower. ✓$_J$Mom criticize-PFV 3SG.

'Xiaoming didn't shower at 7pm. (His) mom criticized him.'

b. ✓$_J$ Xiaoming wanshang-qidian ***bu*** xizao bei Mama piping-le.
✓$_J$ Xiaoming night-7.o'clock NEG.b shower PASS Mom criticize-PFV.

'Xiaoming didn't shower at 7pm and was criticized by mom.'

c. Xiaoming shangge-yue ***bu*** xizao. ✓$_J$Mama piping-le ta.
Xiaoming up.CL-month NEG.b shower. ✓$_J$Mom criticize-PFV 3SG.

'Xiaoming didn't shower last month. (His) mom criticized him.'

d. ✓$_J$ Xiaoming shangge-yue ***bu*** xizao bei Mama piping-le.
✓$_J$ Xiaoming up.CL-month NEG.b shower PASS Mom criticize-PFV.

'Xiaoming didn't shower (at some point) last month and was criticized by mom.'

1.2.3 Gang ... Jiu ...

A further contrast between *mei(-you)* and *bu* comes from *gang...jiu...* ('just...then...'), a grammaticalized construction that encodes event sequencing: the event in the second clause must immediately follow the event in the first.

6. Lü-deng ***gang*** liang, A-Ming ***jiu*** cai-xia-le youmen.
green-light just bright, A-Ming then step-down-PFV accelerator.

'(Immediately after) the moment the light turned green, A-Ming hit the gas.'

Two properties show that *gang...jiu...* is strictly restricted to this event sequencing reading. First, it does not allow an imperfective viewpoint in either the antecedent clause or the anaphor clause:

7. Lü-deng ***gang*** (*zhengzai) liang, A-Ming ***jiu*** (*zhengzai) cai-xia-le
green-light just (*PROG) bright, A-Ming then (*PROG) step-down-PFV
youmen.
accelerator.

'The moment the light turned green, A-Ming hit the gas.'

Second, stative eventualities in either clause must take a perfective marker (such as *le* or *wan*), or be coerced into an inchoative reading. This is expected under narrative sequencing, since only bounded events advance the reference time. In the examples below, for instance, stative *zhidao* ('know') is interpreted as 'come to know' in both the antecedent and the anaphor clause:

8. a. Jingcha ***gang*** **zhidao** xiongshou de mingzi, Laowang ***jiu*** chong-chu-le men.
police just **know** murderer POSS name, Laowang then rush-out-PFV door.

'The moment the police **came to know** the murderer's name, Laowang rushed out the door.'

b. Laowang ***gang*** chong-chu-le men, jingcha ***jiu*** **zhidao** xiongshou de mingzi.
Laowang just rush-out-PFV door, police then **know** murderer POSS name.

'The moment Laowang rushed out the door, the police **came to know** the murderer's name.'

Since *gang...jiu...* encodes grammaticalized event sequencing reading, it offers a direct test of how each negator behaves. The two negators diverge here: my consultants and I find *bu* natural in this construction but *mei(-you)* degraded:

9. a. Ban li de tongxue dou zai zuo zuoye. Xiaoming ***gang*** **bu** zuo
class in POSS classmate all PROG do homework. Xiaoming just NEG.B do
zuoye, laoshi ***jiu*** zoujin-le jiaoshi.
homework, teacher then walk.into-PFV classroom.

'Everyone in the class was doing homework. The moment Xiaoming refused to do his, the teacher walked into the classroom.'

?* b. Ban li de tongxue dou zai zuo zuoye. Xiaoming ***gang*** **mei(-you)** zuo
class in POSS classmate all PROG do homework. Xiaoming just NEG.M do
zuoye, laoshi ***jiu*** zoujin-le jiaoshi.
homework, teacher then walk.into-PFV classroom.

'Everyone in the class was doing homework. The moment Xiaoming didn't do his, the teacher walked into the classroom.'

	English Sentential *not*	*mei(-you)*	*bu*
Narrative, no time frame	✓	✓	✓
Narrative, small time frame	✓	✓	✓
Narrative, large time frame	?	?	✓
gang...jiu...	—	?*	✓

Table 1: Availability of the 'just-after' reading with negation.

1.3 Summary and Paper Organization

Table 4 summarizes these patterns: *mei(-you)* behaves like English *not*, while *bu* behaves differently. In the rest of this paper (particularly Section 4), I will further show that *mei(-you)* is the Mandarin reflex of *not*, both syntactically and semantically, while *bu* has no counterpart in English.

I argue that these observations reflect special properties of negation itself: negation in discourse can introduce negative events, in the sense of Bernard & Champollion (2018, 2023), which can then enter into anaphoric dependencies. Building on their work, I formulate a Temporalized Principle of Negation (TPN), from which semantic inference derives both the anaphoric accessibility of negated events and the interaction between negation and tense-aspect.

I also argue that the two Mandarin negators reach negative events differently: *mei(-you)*, like *not*, encodes standard sentential negation and yields a negative event reading via TPN inference, whereas *bu* directly encodes negative eventualities.

The rest of this paper is organized as follows. Section 2 introduces the accessibility constraint on negation, reviews previous studies, and shows why their limitations call for a different conception of negation in discourse. Section 3 turns to Mandarin two negators. Section 4 introduces negative eventualities and develops a temporal-semantic implementation of Bernard & Champollion (2023). Section 5 explains how the tool may help with explaining our data.

2 Negation, Event Anaphora, and Previous Approaches

2.1 Event Anaphora and the Accessibility Constraint

The role of events in the progression of narrative discourse has been extensively discussed in the dynamic semantics literature, including Partee (1973, 1984), Kamp and Reyle (2013), and Asher and Lascarides (2003, 2007), etc.

In Kamp and Reyle's (2013) framework, a newly introduced event is placed in a direct temporal relation with its antecedent event. In the alternative approach like Partee (1984), Hinrichs (1986), Dowty (1986), the anaphoric dependency is mediated by a reference time/topic time. The two approaches differ only in minor technical details and share the same core intuition. In this paper, I follow Partee and assume that event anaphora is time-mediated: the subsequent event is located with respect to a time that is placed *just after* the antecedent event.

Discourse anaphora is closely related to accessibility constraint. Dynamic theories, despite their various model-theoretic implementations, share the common conception of meaning as information update. The idea of accessibility is present in one form or another in Discourse Represenation Theory (DRT; Kamp, 2013; Kamp & Reyle, 2013), Dynamic Predicate Logic (DPL; Gronendijk & Stokhof, 1991), Compositional Discourse Representation Theory (CDRT; Muskens,

1994), and many other dynamic logics.

(In-)accessibility under negation is entailed by their fundamental mechanisms. This extends naturally to event anaphora. Any attempt to preserve event accessibility – say, by posting a hybrid negator that only state-tests/blocks *e*-type (entity) variables but not *v*-type (event) ones [6] – would derive contradictions such as "No one cried, but some crying events occurred."

2.2 Krifka's Fusion Based Negation and De Swart & Molendijk (1999)

The tension between event anaphora and negation is now apparent. While negation systematically blocks anaphoric access, negated events can nevertheless participate in narrative discourse under suitable timeframes. This observation is not new. The peculiar property of negated entities that they can sometimes be referred to and quantified over has long been noted in the literature. The type of anaphoric accessibility that concerns us here can be viewed as a more specific instance of this broader phenomenon.

A wide-ranging treatment of such cases was developed in Krifka's (1989) event algebra, commonly known as fusion-based negation. This approach was later adopted by De Swart and Molendijk (1999) to account for event anaphora under negation within the DRT framework.

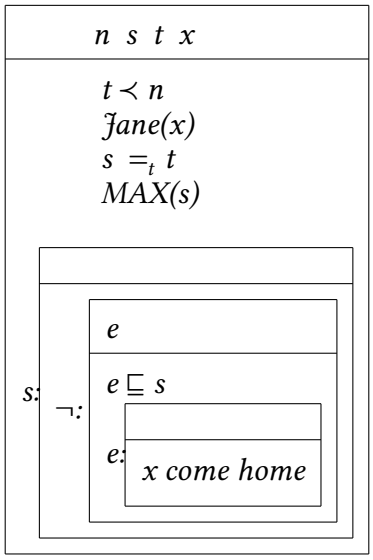
Krifka defines a *maximal event* as the fusion of all the events in a given time interval. His definition relies on a time-trace function τ and a sub-interval relation $\sqsubseteq_T$ between times:

$$\forall e.MAX(e) \iff \exists t.e = \bigsqcup(\lambda e.\tau(e) \sqsubseteq_T t)$$

For Krifka, the negation of a predicate P denotes some maximal event that contain no P-event as part:

$$[\![not]\!] = \lambda P_{vt}.\lambda e.MAX(e) \wedge \neg\exists e'.P(e') \wedge e' \sqsubseteq e$$

De Swart and Molendijk (1999) incorporate this treatment into DRT. A negative sentence such as 'Jane didn't come home' is then represented as



"Jane didn't come home."

[6] See endnote [1]

where the discourse referent s denotes a maximal event, and n denotes the speech time. This representation says that a maximal event s occurred at a past time t, and that no event of Jane's coming is included as a part of s. As De Swart and Molendijk points out, negation thus introduces an accessible event referent s that can serve as an antecedent for subsequent discourse updates. However, there is reason to think that this approach is not on the right track.

The argument comes from what I call the *same-time problem*. Consider the discourse in (5):

10. [1] On Sunday, John did not do his homework.
    [2] His brother Tom did not clean the bedroom.
    [3] John's not doing homework made his mom upset.
    [4] Tom's not cleaning bedroom did not make mom upset.

(5.[1]) and (5.[2]) both spans the entire Sunday. Under the fusion-based negation, each negated sentence introduces a maximal event corresponding to the fusion of all other events occurring on Sunday. As a result, the two negated clauses end up introducing the same negative event.

This leads to a contradiction: the very same maximal event has made mom upset and made her not upset.

A **second** argument comes from time-frame adverbs. As already noted in Section 1, the contrast between (2.a) and (2.b) shows that the size of the temporal frame affects the felicity of negated events in narrative discourse. This sensitivity is not captured by De Swart and Molendijk. In principle, their system allows one to fuse all non-P events within a large interval such as *June* in the case of (2.b), thereby predicting the felicity of an event anaphor.

The **final** argument concerns lexical aspect. As discussed in Krifka (1989) and De Swart & Molendijk (1999), maximal events have stative properties. A fusion-based analysis therefore predicts that negated sentences in narratives should be stative.

A semantic test shows that they are not. The 'happen' test, adopted from Fábregas and González Rodríguez (2020), distinguishes the two classes: eventive sentences can be picked up by a "this happened when ..." continuation, whereas stative sentences cannot:

11. ✓a. John spoke an English sentence. This happened when ...
    #b. John knew English. This happened when ...

English negative sentences in plain narratives pass the test, showing that they have an eventive reading rather than a stative one:

12. The thief did not leave (and this happened when Henry told him to). Henry shot him.

Mandarin *fasheng* ('happen') continuation draws the same distinction between events and states:

13. ✓a. A-hong shuo-le yi-ju yingyu. Zhe **fasheng** zai ...
    A-hong speak-PFV one-CLF English. This **happen** at ...
    'A-hong spoke an English sentence. This happened when ...'

    #b. A-hong hui shuo yingyu. Zhe **fasheng** zai ...
    A-hong MOD speak English. This **happen** at ...
    'A-hong can speak English. This happened when ...'

In Mandarin narratives without time-frame adverbs, however, a *fasheng* continuation is available with both *mei(-you)* and *bu*:

14. Zhangsan { ✓***mei(-you)*** / ✓***bu*** } zhan-qilai (zhe fasheng zai laoshi rang ta
Zhangsan { ✓NEG.m / ✓NEG.b } stand.up (this happen at teacher let/cause 3SG
zhan-qilai de shihou). Laoshi piping-le ta.
stand.up DE time). teacher criticize-PFV 3SG.

'Zhangsan did not stand up (and this happened when the teacher asked him to stand up). The teacher criticized him.'

Since maximal events are stative, these results argue against treating negated sentences in plain narratives as introducing maximal events.

2.3 Intensionality?

Hofmann (2019, 2022) discusses how modality can license **nominal** anaphora under negation:

15. ✓ There is no bathroom. It would be easier to find.

Although Hofmann does not extend this approach to event anaphora, it suggests a possible account of the present data, since counterfactual events also seem accessible under negation in English:

16. a. The thief did not resist. John would shoot him.
b. The thief did not resist. (But if he did,) John would shoot him.

An intensional account seems especially relevant for Mandarin *bu*, which is often argued to carry a covert volitional modal. Huang (1988) observed that (17a) and (17b) have similar meanings and proposes a null volitional modal head for (17a), as in (17c):

17. a. Zuo-wan A-Qiu bu xie zuoye.
yesterday-night A-Qiu NEG.B write homework.
'Last night A-Qiu refused to do homework.'

b. Zuo-wan A-Qiu **xiang** bu xie zuoye.
yesterday-night A-Qiu **want** NEG.B write homework.
'Last night A-Qiu wanted to not do homework.'

c. Zuo-wan A-Qiu $\emptyset_{\text{MOD}}$ bu xie zuoye.
yesterday-night A-Qiu $\emptyset_{\text{MOD}}$ NEG.B write homework.
'Last night A-Qiu wanted to not do homework.'

However, I do not analyze the event anaphora cases of interest as involving intensionality, for two reasons. **First**, the shooting in (16a.) need not occur in the actual world and is overtly modalized, whereas the shooting in the plain narrative in (1) is an actual event described by an episodic clause. **Second**, an intensional account offers no straight-forward explanation of why event anaphora should be sensitive to the size of the time frame.

For Mandarin, I likewise argue that event anaphora does not involve intensionality under either *mei(-you)* or *bu*. There are again two reasons. **First**, a volitional account of *bu* does not extend to *mei(-you)*, which also allows event anaphora in plain narratives and with small time frames, yet to the best of my knowledge has never been analyzed as volitional.

Second, it is not even clear that *bu* in narratives encodes covert volition at all. If it did, plain *bu* should pattern with overtly modalized *bu xiang* ('not want') in surviving contradictions, but it does not:

18. ✓a. A-ming **bu xiang** xie zuoye, dan ta haishi xie-le.
A-ming **NEG.B want** write homework, but 3SG still write-PFV.
'A-ming didn't want to write the homework, but he wrote it anyway.'

    b. A-ming **bu xie zuoye, dan ta haishi xie-le.
A-ming **NEG.B** write homework, but 3SG still write-PFV.
'A-ming didn't write the homework, but he wrote it anyway.'[7]

    ✓c. Laoshi rang Xiaoming zhan qilai. Xiaoming ***bu xiang*** zhan, dan zhan-le qilai.
Teacher ask Xiaoming stand up. Xiaoming **NEG.b want** stand, but stand.PFV up.
'The teacher asked Xiaoming to stand up. Xiaoming didn't want to, but still stood up.'

    d. Laoshi rang Xiaoming zhan qilai. Xiaoming ***bu* zhan, dan zhan-le qilai.
Teacher ask Xiaoming stand up. Xiaoming **NEG.b** stand, but stand.pfv up.
'The teacher asked Xiaoming to stand up. Xiaoming didn't stand up but still stood up.'

2.4 Interim Summary

	English Sentential *not*	*mei(-you)*	*bu*
'Just-after' reading			
Narrative, no time frame	✓	✓	✓
Narrative, small time frame	✓	✓	✓
Narrative, large time frame	?	?	✓
gang...jiu...	—	?*	✓
Aktionsart			
Eventive under 'happen' test (narratives without time-frame)	✓	✓	✓
Modality			
Intensional in narratives	✓	✓	✓

Table 2: Properties of English *not*, Mandarin *mei(-you)*, and Mandarin *bu*.

[7](18b,d) are severely degraded, as the second clauses clearly contradict the first.

3 The Syntax of *Bu* and *Mei* (and comparing to English)

In this section, I show one further syntactic difference between Mandarin *bu* and *mei(-you)*, and compare them to English sentential *not*: *bu* is syntactically lower than *mei(-you)*. It can occur as low as inside *v*P, where it is best analyzed as a *v*P adjunct.[8] *Mei(-you)*, however, predictably stays in the TMA domain, just like English sentential *not*.

3.1 *Bu* is Syntactically Lower than *Mei*

The first and most direct piece of evidence is that *mei(-you)* and *bu* can in fact co-occur, in which case *mei(-you)* must precede *bu*:

19. ✓a. Wo ***mei(-you) bu*** qu shangxue.
1SG NEG.M NEG.B go attend.school.
'I didn't NOT go to school.'

b. Wo ***bu* ***mei(-you)*** qu shangxue.
1SG NEG.B NEG.M go attend.school.
Intended: 'I didn't NOT go to school.'

If one follows Kayne's (1994) Linear Correspondence Axiom, on which linear order reflects asymmetric c-command, the fixed order in (19) indicates that *mei(-you)* must be structurally higher than *bu*.

There is further evidence showing that *bu* can occur within the *v*P domain, whereas *mei(-you)* cannot. The first argument comes from *v*P pseudo-clefts. In Mandarin, a *v*P can appear before the copula *shi*, followed by a relative clause headed by *shi* ('thing'):

20. a. Zhangsan hui xie daima.
Zhangsan MOD write code.
"Zhangsan is able to write codes".

b. [Xie daima] shi [Zhangsan hui zuo de shi]$_{\text{REL}}$.
write code COP Zhangsan MOD do REL thing.
"Writing codes is what Zhangsan can do."

Notably, as in English, the clefted constituent cannot contain outer-aspectual or modal morphemes, which supports that it is a *v*P:

21. **No Asp**

a. Zhangsan zai xie daima.
Zhangsan PROG write code.
"Zhangsan is writing codes."

b. [(*zai) Xie daima] shi [Zhangsan (✓zai) zuo de shi]$_{\text{REL}}$.
(*PROG) write code COP Zhangsan (✓PROG) do REL thing.
"Writing code is what Zhangsan is doing".
" *Being writing code is what Zhangsan does."

[8] Previous work has shown that *bu* can also occur higher, in the Tense-Modal-Aspect (TMA) domain. This paper concerns only the lowest *bu*, and I remain neutral on whether Mandarin has multiple *bu*'s.

22. **No Mod**

    a. Lisi keyi jvjia bangong.
    Lisi MOD at.home work.

    "Lisi can work at home."

    b. [(*keyi) Jvjia bangong] shi [Lisi (✓keyi) zuo de shi]$_{\text{REL}}$.
    (*MOD) at.home work COP Lisi (✓MOD) do REL thing.

    "Working at home is what Lisi can do."
    " *Being able to work at home is what Lisi does."

My consultants and I find that *bu* must appear in the clefted constituent, whereas *mei(-you)* is most natural inside the relative clause:

23. What did John do last night to make his mom so angry?

    ✓a. Ta zuowan bu xie zuoye.
    3SG yesterday.night NEG.B write homework.
    'He didn't do his homework last night.'

    b. [(✓***bu***) xie zuoye] shi [ta zuowan (****bu***) zuo de shi]$_{\text{REL}}$.
    (✓NEG.B) write homework COP 3SG yesterday.night (*NEG.B) do REL thing.

    "Not doing homework is what he did last night."

    ✓c. Ta zuowan mei(-you) xie zuoye.
    3SG yesterday.night NEG.M write homework.
    'He didn't do his homework last night.'

    d. [(?****mei(-you)***) xie zuoye] shi [ta zuowan (✓***mei(-you)***) zuo de shi]$_{\text{REL}}$.
    (?*NEG.M) write homework COP 3SG yesterday.night (✓NEG.M) do REL thing.

    "Not doing homework is what he did last night."

This supports an analysis in which *bu* occurs within the *v*P domain while *mei(-you)* occupies the TMA domain. It also independently supports the point that low *bu* does not contain a covert modal: since modals cannot appear in the clefted constituent.

A second piece of evidence for low *bu* comes from its scope relative to low modals. Mandarin has modals that sit syntactically low, close to *v*P (cf. the cartographic hierarchy in Cinque, 1999), such as *xiang* ('want to') and *neng* ('be able to'):

24. A-Hong xiang tan jita.
    A-Hong MOD$_{\text{Vol}}$ play guitar.

    "A-Hong wants to play guitar".

Mei(-you) obligatorily scopes over the low volitional modal *xiang*, while *bu* can scope either over or under it:

25. a. A-Hong (✓mei(-you)) xiang (*mei(-you)) tan jita.
    A-Hong (✓NEG.M) MOD$_{\text{Vol}}$ (*NEG.M) play guitar.

    'A-Hong didn't want to play guitar.'

b. A-Hong (✓bu) xiang (✓bu) tan jita.
A-Hong (✓NEG.B) MOD_{Vol} (✓NEG.B) play guitar.

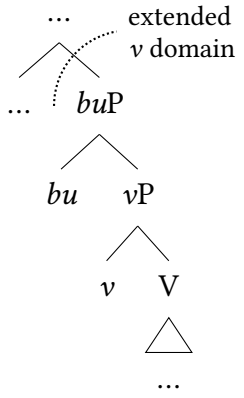
bu > *xiang*: 'A-Hong doesn't want to play guitar.'
xiang > *bu*: 'A-Hong wants to not play guitar.'

Some other evidence comes from the negators' positions relative to aspectual morphemes and from *v*P ellipsis, which I set aside here for reasons of space.

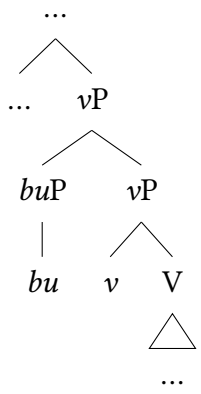
3.2 *Bu* as a *v*P-adjunct

The previous section showed that *bu* can occur low, within the *v*P domain. Two analyses are possible. In the first, *bu* heads its own phrase, selects *v*P, and extends the *v*P domain (26a). In the second, *bu* is simply a *v*P adjunct (26b).

26.



(a) *bu* as a head selecting *v*P



(b) *bu* adjoins to *v*P

I argue that we should adopt the adjunction analysis. Supporting data comes from *v*P coordination and scope ambiguity.

27. a. Zhangsan zongshi [$_{\text{coor}}$[$_{v\text{P}}$sahuang] he [$_{?}$bu xie zuoye]], laoshi jiazhang
Zhangsan often lie CONJ NEG.B write homework, teacher parents
dou na ta mei banfa.
all take 3SG NEG.M idea.

'Zhangsan always lies and doesn't do his homework. His teachers and parents can't do anything about him.'

b. LaoLi tiantian [$_{\text{coor}}$[bu chi wufan] he [chouyan]]. Ta laopo hen danxin.
Old.Li day.day NEG.B eat lunch CONJ smoke.cigarette. 3SG wife DEG worry.

'Old Li skips lunch and smokes every day. His wife is very worried.'

c. LaoLi tiantian bu [$_{\text{coor}}$chi wufan he chouyan]. Ta zhi gu gongzuo.
Old.Li day.day NEG.B eat lunch CONJ smoke.cigarette. 3SG only mind work.

'Every day, Old Li neither eats lunch nor smokes. He only cares about work.'

In (27a), the *v*P [*sahuang*] ('lie') is coordinated with [*bu xie zuoye*] ('not do homework'). On the common assumption that only constituents of the same category can be coordinated (Chomsky, 1957; Schachter, 1977; Williams, 1978; among others), [*bu xie zuoye*] must also be a *v*P. It follows that *bu* does not project its own label, which is a typical property of adjuncts.

Moreover, the string [*tiantian bu chi wufan he chouyan*] in (27b,c) is ambiguous, and the follow-up sentences in (27b) and (27c) select each reading: in (27b), Old Li skips lunch but does smoke, while in (27c), he neither eats lunch nor smokes. This ambiguity follows naturally if *bu* can adjoin either to the whole coordination, giving it scope over both conjuncts, or to the first *v*P alone, giving it scope over that conjunct only. By contrast, if *bu* is a head selecting *v*P, the same-category requirement allows only the wide-scope reading, since [*bu*P *bu chi wufan*] could not coordinate with the *v*P *chouyan*. I therefore treat *bu* as a *v*P adjunct in what follows.

3.3 Interim Summary

The syntactic properties of the two negators can now be added to the table, alongside English *not*:

	English Sentential *not*	*mei(-you)*	*bu*
'Just-after' reading			
Narrative, no time frame	✓	✓	✓
Narrative, small time frame	✓	✓	✓
Narrative, large time frame	?	?	✓
gang...jiu...	—	?*	✓
Aktionsart			
Eventive under 'happen' test (narratives without time-frame)	✓	✓	✓
Modality			
Non-intensional in narratives	✓	✓	✓
Syntactic Position			
*v*P Domain (low)	✘	✘	✓
TMA Domain (high)	✓	✓	(✘)

Table 3: Properties of English *not*, Mandarin *mei(-you)*, and Mandarin *bu*.

4 A Temporal Semantics for *Neg* and Event Anaphora

Two important properties of negation in English and Mandarin narratives emerge from the discussion so far: (1) it is eventive, but its events are not 'fused'; and (2) it is non-intensional. In this section, I develop a formal tool that can capture these properties. It can be seen as a temporal-semantic extension of Bernard & Champollion (2023; henceforth B&C). While B&C provide a well-motivated alternative to the fusion-based approach, their proposal is not temporalized and therefore cannot be directly applied to discourse update.

The core intuition is simple: negation can introduce events, but these events are not derived by summing all events within a given time interval.

This section proceeds in two steps. First, I recast B&C's insights in temporal-semantic terms. Second, I demonstrate that this enriched tool accounts for the observed patterns of event anaphora in both English and Mandarin.

4.1 Briefly Introducing B&C

B&C propose that natural language negation introduces negative events, which are not ontologically distinct from their ordinary (positive) counterparts. Whether an event counts as negative or positive depends on its description: any event in D_v can be viewed as negative under a negated description or positive under a non-negated one. For example, the same event can be positive under "John stayed" but negative under "John did not leave".

Formally, B&C define a function *Neg* of type $\langle vt, vt\rangle$, which maps an event predicate $\lambda e.P(e)$ to a predicate $\lambda e.Neg(P)(e)$ that holds of negative P-events, i.e., events whose occurrence is incompatible with the occurrence of any P-event. In this sense, every event predicate has its own set of negative events.

The guiding intuition is that, for a given description, either a P-event or a negative P-event occurs, but not both. Thus, if no event of John's leaving occurs, a negative event of leaving (e.g., a staying event) occurs, and vice versa. Formalizing this intuition requires existential quantification. Crucially, however, B&C point out that it cannot be stated as a simple equivalence such as

$$\neg\exists e.P(e) \iff \exists e.Neg(P)(e) \quad \text{✘}$$

because existential quantification ranges over the entire event domain. If P has an empty extension, as the left-hand side requires, it provides the same input to *Neg* (the empty set) regardless of its descriptive content. Distinct predicates with empty extensions would therefore yield the same negative events. This incorrectly predicts that sentences such as "John did not leave" and "The thief did not put down his gun" involve identical negative events.

B&C therefore distinguish between event existence and event occurrence. They posit a predicate $\lambda e.actual(e)$, which characterizes the set of events that occurred, and propose an axiom relating negative events to negation:

Principle of Negation

$$\forall P_{vt}.\ \big[\exists e.\ P(e) \wedge actual(e) \iff \neg\exists e'.\ Neg(P)(e') \wedge actual(e')\big],$$

That is, some P-event occurred if and only if no negative P-event occurred. The axiom thus guarantees both that positive and negative P-events cannot both occur and that at least one of them must.

This approach has several advantages:

- It is non-mereological: negative events are not sums of events within an interval, so it avoids the same-time problem (Section 2.2).
- It accounts for the Aktionsart of negated predicates, which behave as eventive.
- It avoids radical ontological commitments, since negative events are ordinary events under a particular description.

4.2 Temporal-Semantic Formalization

As pointed out at the beginning of this section, B&C's proposal offers a novel approach to negation and negative events, but it is atemporal and thus cannot be directly applied to discourse update or related phenomena. This subsection offers a temporal-semantic implementation of it.

4.2.1 Time-event Ontology and Homomorphism

Following common practice, assume a time structure T and an event structure E, with a time domain D_i and a sufficiently rich event domain D_v. Times and events are equipped with binary mereological sum operations $\oplus_T$: $D_i \times D_i \to D_i$ and $\oplus_E$: $D_v \times D_v \to D_v$. These generalize to a sum operation $\bigsqcup$ over non-empty sets, including infinite ones, provided that all members fall within some common upper bound. As usual, a time-trace function τ maps events to their run times and is homomorphic with respect to these sums, (28a). The part-of and proper-part-of relations are defined based on the sum operations, (28b–c). Together, these assumptions guarantee that if an event is part of another event, its run time is part of the other's run time:

28. a. $\forall e, e'.\ \tau(e \oplus_E e') = \tau(e) \oplus_T \tau(e')$; more generally, $\tau(\bigsqcup P) = \bigsqcup\{\tau(e) \mid e \in P\}$
    b. $\forall x, y.\ x \leq y \iff x \oplus y = y$
    c. $\forall x, y.\ x < y \iff x \leq y \wedge x \neq y$

A temporal precedence relation $\prec_T$ is assumed to be transitive and irreflexive (and hence asymmetric), and time is assumed to have no end: $\forall t \exists t'.\ t \prec_T t'$. For any $t \in D_i$, if t has no proper parts, then t is a moment (i.e., an atom in the semi-lattice). For any two non-overlapping convex intervals, one must precede the other; hence, moments are linearly ordered by $\prec_T$.

4.2.2 Occurance, Existence, and Event Actualization

An event e is said to occur relative to a time t iff $\tau(e) \leq_T t$, i.e., e is within t's time span. On this basis, define a temporal actualization predicate *Actual* of type $\langle i, \langle v, t\rangle\rangle$ as (29):

29. Actualization in times: $\forall t, e.\ \mathit{Actual}(t)(e) \iff \tau(e) \leq_T t$.

Assume further that there are no empty times (i.e., times at which no event occurs)[9]:

30. No empty times: $\forall t.\ \exists e.\ \tau(e) \leq_T t$

[9]Different strategies for ruling out empty times are discussed in Krifka (1998). Alternatively, one may adopt an analysis such as Kamp (1979), where times are non-primitive elements defined via primitive events. Under that analysis, the absence of empty times is automatically guaranteed.

Since every time is followed by a later time, and some event occurs at that later time, no time contains all events: $\forall t.\exists e.\neg(\tau(e) \leq_T t)$. This yields a principled distinction between event existence (in D_v) and event occurrence (in time): the set of events that exist is D_v, while those occurring relative to any t form a proper subset $\{ e \mid \tau(e) \leq_T t \} \subset D_v$. Thus, consistent with B&C, non-occurrence does not imply non-existence.

4.2.3 Temporalized Principle of Negation (TPN)

Following B&C, I assume a function that maps an event predicate to its corresponding negative events, but I relativize it to the time index of the interpretation function:

$$[\![Neg]\!]^t = \lambda P_{vt}.\lambda e.\ Neg_t(P)(e)\ :\ \langle vt, vt\rangle$$

Under these assumptions, the Principle of Negation can be reformulated as the following Temporalized Principle of Negation (TPN):

31. **Temporalized Principle of Negation**
$\forall t, P_{vt}.\ \left[\neg\exists e.\ P(e) \wedge Actual(e)(t) \iff \exists e'.\ Neg_t(P)(e') \wedge Actual(e')(t)\right]$

TPN states that if no P-event occurs at a time t, then a negative P-event relative to t occurs at t,[10]. and conversely. [11]

4.2.4 Logical Properties of TPN

Important logical properties are preserved. As expected, Double Negation Cancellation (DNC) is validated: the occurrence of a negative-negative-P event at t entails the occurrence of some P-event at t, and vice versa. The proof proceeds as in B&C:

32. **Proof of Double Negation Cancellation**
Given a time t, and a predicate P,
$\exists e.\ Neg_t(Neg_t(P))(e) \wedge Actual(e)(t)$
$\iff \neg\exists e.\ Neg_t(P)(e) \wedge Actual(e)(t)$
$\iff \neg\neg\exists e.\ P(e) \wedge Actual(e)(t)$
$\iff \exists e.\ P(e) \wedge Actual(e)(t)$

B&C require actualization to satisfy certain algebraic properties in order for event negation to obey the Laws of Excluded Middle and Non-Contradiction (pp. 597–598). They posit the following axioms:

33. a. Distributivity of actualization: $\forall e.\ Actual(e) \rightarrow \forall e' \leq_E e.\ Actual(e')$
b. Cumulativity of actualization: $\forall P.\ [\forall e \in P.\ Actual(e)] \rightarrow Actual(\bigsqcup P)$

---

[10] The time index is essential. *Actual* is upward monotonic: an event that is actual at t is also actual at any larger time. If *Neg* were not time-dependent, a negative P-event at a P-free time t_1 would still be actual at a larger time t' containing a P-event, so a P-event and a negative P-event would both occur at t' (Compare B&C's modal variant, in which negative events can vary (p. 596)).

[11] It thereby entails (30), since either a P-event or a negative P-event occurs at any time. I nevertheless state (30) as a separate assumption, because the distinction between event existence and event occurrence is a general property of times and events, and should not depend on the semantics of negation

Axiom (33a) states that every part of an actual event is itself actual, while (33b) states that the sum of actual events is actual.

In the temporal extension, both properties follow as theorems from the time-event homomorphism: if e occurs relative to t, all its parts do too, (34a), and the sum of any non-empty set of events occurring in t also occurs in t, (34b):

34. a. Distributivity of temporal actualization: $\forall t, e.[\tau(e) \leq_T t \rightarrow \forall e' \leq_E e.\tau(e') \leq_T t]$
    b. Cumulativity of temporal actualization: for non-empty P, $\forall t.\ [\forall e \in P.\ \tau(e) \leq_T t] \rightarrow \tau(\bigsqcup P) \leq_T t$[12]

Finally, since TPN is a biconditional, it directly yields counterparts of the Laws of Excluded Middle and Non-Contradiction at every time: for any P and t, either a P-event or a negative P-event relative to t occurs at t, but not both.

4.3 Negative Events and Event Anaphora in Narrative Discourse

TPN can now be used to account for event anaphora in discourse. To see how, consider first the syntax and semantics of tense and aspect.

I adopt the view, standard since Reichenbach (1947), Klein (2013), Kratzer (1998), and much subsequent work, that viewpoint (grammatical) aspect is a relation between an event's run time $\tau(e)$ and the topic time t_{Top}. On this approach, perfective aspect requires the event to be temporally contained within the topic time, whereas imperfective aspect requires the topic time to be contained within the event's run time. Formally:

35. a. $PERF_{\langle vt,it\rangle} \stackrel{\text{def}}{=} \lambda P_{vt}.\lambda t_{Top}.\exists e.P(e) \wedge \tau(e) \leq_T t_{Top}$
    b. $IMPERF_{\langle vt,it\rangle} \stackrel{\text{def}}{=} \lambda P_{vt}.\lambda t_{Top}.\exists e.P(e) \wedge \tau(e) \geq_T t_{Top}$

The English simple past is assumed to default to perfective viewpoint. For instance, "Mary wrote a letter" (telic) entails completion of the letter-writing event, and "Mary watched TV" (atelic) entails the occurrence of a temporally bounded TV-watching event. The English progressive, by contrast, is taken to instantiate imperfective aspect: "Mary was writing a letter" does not entail that the letter was finished.[13]

Now consider narrative discourses involving negated events in the simple past:

36. a. [1] Henry told the thief to leave. [2] The thief did not leave. [3] Henry shot him.

r_0	r_1	r_2
e_0	e_1	e_2

    b. [1] John invited all his friends. [2] They didn't show up. [3] John decided to go out into the street and bring in all the homeless in the neighbourhood.

(adapted from De Swart & Molendijk 1999: 7)

[12] This approach potentially bears on the imperfective paradox, if the paradox is taken to concern event occurrence rather than event existence.

[13] The English progressive has also been argued to involve event-type shifting, mapping activities or accomplishments onto states. For this reason, some alternative approaches analyze the progressive as a stativizer (cf. Vlach 1981; Moens 1987; Kamp & Reyle 2013).

In both (36a) and (36b), the negated clause is fully integrated into the narrative progression.

Following Partee (1984) and Hinrichs (1986), I assume that narrative progression is driven by a basic update rule, on which the topic time of each clause is definite: it is the reference time updated by the preceding event.

37. **Basic Discourse Update**
Given a clause interpreted at the current reference time r_n, if its logical form introduces an event e with $\tau(e) \leq_T r_n$:

    a. e is added to the accessible discourse referents; and

    b. the reference time is updated to r_{n+1}, where $\tau(e) \prec\!\!\prec_T r_{n+1}$.

A negated clause poses a problem for this rule. The logical form of clause [2] in (36a) contains only $\neg\exists e_1.\, Leave(\iota(thief), e_1) \wedge \tau(e_1) \leq_T r_1$, which introduces no event. Rule (37) therefore predicts that this clause provides no antecedent and does not update the reference time. Yet clause [3] is understood as following, and responding to, the thief's not leaving. To resolve this, I propose that the TPN inference applies as a last-resort operation:

38. **TPN Update (last resort)**
Let ϕ be the logical form of a negated clause, and let ψ be a subformula of ϕ of the form $\neg\exists e.\, P(e) \wedge \tau(e) \leq_T t$. ϕ is updated to

$$\phi\big[\psi \mapsto \exists e'.\, Neg_t(P)(e') \wedge \tau(e') \leq_T t\big],$$

where e' is a new discourse referent, iff:

    a. **Trigger:** the discourse requires an event antecedent located within t, and no accessible event e'' satisfies $\tau(e'') \leq_T t$; and

    b. **In situ:** ψ occurs in ϕ itself.

To see how this works, consider clause [2] of (36a), whose topic time is the reference time r_1 immediately following Henry's telling event e_0. It is represented as

$$[\tau(e_0) \prec\!\!\prec_T r_1] \wedge [r_1 \prec_T s^*] \wedge \underline{\big[\neg\exists e_1.\, Leave\big(\iota(thief), e_1\big) \wedge \tau(e_1) \leq_T r_1\big]},$$

which states that no leaving event by the thief occurred within r_1, where $\prec\!\!\prec_T$ means 'just after'. Crucially, the underlined conjunct has exactly the same logical form as the left-hand side of TPN. Since no accessible event lies within r_1 (the telling event e_0 precedes it) and the conjunct occurs in situ, (38) applies, yielding:

$$\tau(e_0) \prec\!\!\prec_T r_1 \wedge r_1 \prec_T s^* \wedge \big[\exists e_1'.\, Neg_{r_1}\big(\lambda e_1.Leave(\iota(thief), e_1)\big)\big(e_1'\big) \wedge \tau(e_1') \leq_T r_1\big].$$

The event variable e_1' is discourse-accessible: a negative event of the thief's leaving is introduced into the discourse and can serve as an antecedent for subsequent event anaphora. By (37b), it also updates the reference time to r_2, the time of Henry's shooting event in clause [3]. Parallel reasoning applies to clause [2] of (36b) and, more generally, to negated perfective clauses in narrative discourse.

4.4 Advantages

This approach avoids the *same-time problem* discussed in Section 2.2. Since *Neg* can yield distinct negative events, even when these events are located at the same time.

The approach also resolves the time-frame problem illustrated in (2), repeated here as (39):

39. a. ✓$_J$ At noon, Johnny did not see his father. He then cried.
    b. ?$_J$ In June, Johnny did not see his father. He then cried.

The contrast follows from the scope of negation. *At noon* is by default resolved to the closest noon (Kamp & Reyle, 2013), and *at* identifies the topic time with it. TPN update applies at the top level and the negative event is accessible. The first sentence of (39a) is represented as follows, with the topic time identified with the closest noon:

$$noon \prec_T s^* \wedge \left[\neg\exists e_1.\ See(johnny, his\ father, e_1) \wedge \tau(e_1) \leq_T noon\right].$$

Since no accessible event lies within *noon* and the bracketed conjunct occurs in situ, (38) applies, yielding:

$$noon \prec_T s^* \wedge \left[\exists e_1'.\ Neg_{noon}\big(\lambda e_1.See(johnny, his\ father, e_1)\big)(e_1') \wedge \tau(e_1') \leq_T noon\right].$$

By (37), e_1' is now accessible and updates the reference time.

By contrast, (39b) is represented with narrow scope negation as

$$\exists t_{Top}.\ t_{Top} \leq_T June \wedge t_{Top} \prec_T s^* \wedge \left[\neg\exists e.\ See(johnny, his\ father, e) \wedge \tau(e) \leq_T t_{Top}\right].$$

This has a trivial truth condition: it is satisfied as long as there is at least one moment in June at which Johnny does not see his father. The intended reading is instead "At no past time in June did Johnny see his father". To ensure pragmatic informativeness, negation must therefore take wide scope:

$$\neg\exists t_{Top}.\ t_{Top} \leq_T June \wedge t_{Top} \prec_T s^* \wedge \left[\exists e.\ See(johnny, his\ father, e) \wedge \tau(e) \leq_T t_{Top}\right].$$

This is logically equivalent to the following universal statement:

$$\forall t_{Top}.\ \left[t_{Top} \leq_T June \wedge t_{Top} \prec_T s^*\right] \rightarrow \left[\neg\exists e.\ See(johnny, his\ father, e) \wedge \tau(e) \leq_T t_{Top}\right].$$

(38) can still apply in situ to the consequent, yielding the inference that, for every past time in June, a negative event of Johnny's seeing his father occurred. Crucially, however, these negative events are introduced under universal quantification, so they are not accessible from outside it. As a result, the sentence fails to provide an antecedent for event anaphora.[14,15]

[14]Note that the universal does entail statements about particular times, such as June 10, from which a negative event could be inferred. This does not provide an antecedent, however, because accessibility is not closed under entailment (cf. Partee's marble example in Heim, 1982), and (38b) restricts the update to subformulas of the logical form itself.

[15]Some audience members at ICFL-11 noted that this prediction depends on obtaining the appropriate scope for negation, and asked how that scope is guaranteed. I do not have a full answer to this question, and a complete theory of negation scope lies beyond this paper. Since negation scope and pragmatic strength have been extensively studied, I assume that independently motivated mechanisms can derive the correct interpretation.

4.5 Predictions

This approach makes two specific predictions. First, negated **imperfective** sentences should not feed event anaphora in narratives. A negated imperfective clause contains the condition $[\neg\exists e.\ P(e) \wedge \tau(e) >_T t_{Top}]$, which does not have the form required by TPN update, nor does it entail $[\neg\exists e.\ P(e) \wedge Actual(e)(t_{Top})]$: even if no P-event's run time properly contains t_{Top}, there may still be a P-event whose run time falls within t_{Top}.[16] TPN Update therefore does not apply, the clause introduces no event, and by Basic Discourse Update it does not update the reference time.

40. Mary walked into the room. Johnny wasn't doing his homework. Mary got mad.

This aligns with the well-established observation that English progressives function as *Background* (Partee, 1984; Hinrichs, 1986).

Second, negative events introduced in intensional contexts should not be accessible from outside them. (38) can apply within the complement of an attitude verb, but e' is then introduced in the scope of the attitude, so it is inaccessible from outside it:

41. Henry thought that the thief did not leave, and he shot him (*after his not-leaving).

Here, Henry's shooting cannot be interpreted as following the thief's not leaving. By contrast, a continuation that stays within the scope of the attitude should be able to access the negative event, as in modal subordination (Roberts, 1989):

42. Henry thought that the thief did not leave, and that this would anger the guards.

5 Addressing Mandarin Data

5.1 The Proposal for *Mei*: Paralelling English *Not*

	English Sentential *not*	*mei(-you)*	*bu*
'Just-after' reading			
Narrative, no time frame	✓	✓	✓
Narrative, small time frame	✓	✓	✓
Narrative, large time frame	?	?	✓
gang...jiu...	–	?*	✓
Aktionsart			
Eventive under 'happen' test (narratives without time-frame)	✓	✓	✓
Modality			
Non-intensional in narratives	✓	✓	✓
Syntactic Position			
*v*P Domain (low)	✘	✘	✓
TMA Domain (high)	✓	✓	(✘)

Table 4: Properties of English *not*, Mandarin *mei(-you)*, and Mandarin *bu*.

[16] Consider "Mary was not reading from 9 to 10, but she was reading from 9:30 to 9:35".

In Table 3 (copied above) I have shown that Mandarin *mei(-you)* behaves very similarly to English *not* both syntactically and semantically in narrative discourse. I propose that Mandarin *mei(-you)* is just like English sentential negation and can be addressed in the same way as section 4.

5.2 The proposal for *bu*: Restricted *Neg*

Unlike *mei(-you)*, low *bu* can have an eventive reading in practically all narrative contexts, whether with *gang ... jiu ...* or with time frames of different sizes. I have also shown that *bu* can adjoin to *v*P. This suggests that *bu* directly introduces a negative event, rather than obtaining one through TPN.

Since *bu* also carries a reading that the subject volitionally did something, without involving a volitional morpheme, I treat this as an idiosyncratic property of Mandarin low negation and propose the following denotation for *bu*:

$$[\![bu]\!]^{g,t} = \lambda P_{vt}.\lambda e_v.\ [\![Neg]\!]^t(P)(e) \wedge Agent(e) = g(x_i),$$

where x_i is coindexed with the subject of the same clause. The volitional reading arises from this agentivity restriction. Since not every negative event has the subject as its agent, the absence of a P-event does not guarantee a *bu*-event, so *bu* is not equivalent to plain *Neg*.

With perfective aspect, and with the index identified with the topic time, *bu* + *v*P yields:

$$\lambda t.\ \exists e.\ [\![Neg]\!]^t(P)(e) \wedge Agent(e) = g(x_i) \wedge \tau(e) \leq_T t.$$

By TPN, this still entails that no P-event occurred at t. It also explains why negative events are consistently available with *bu*: the event is introduced compositionally, so (38) is not needed. The same holds for the contrast with *mei(-you)* under large time frames. Because *bu* asserts the existence of an agentive negative event, narrow scope is informative even with *in June*. Nothing forces wide scope, so no universal arises, and the event remains accessible.

Other papers to cite and discuss:
Lin (2006)
Lin (2003)
Lee & Pan (2001)
Ernst (1995)
Oshima (2025)
De Swart (1998)

Notes

[1]A version of this idea has been suggested in informal discussion. Indeed, a sentence like "Josef did not see a unicorn" may admit an interpretation as $\exists e.SEE(e) \wedge Ag(e, john) \wedge \neg\exists x.Unicorn(x) \wedge Theme(e, x)$, which says that a seeing event occurred but it was not a unicorn that was seen. In this case, the surface wide-scope negation allows the event variable to remain accessible. However, this interpretation is better attributed to focus semantics, where negation (as a focus operator) targets the F-marked constituent (cf. Jackendoff 1972):

Josef did *not* see [a unicorn]$_F$.

Under this view, the negation does not behave as normal sentential operator. It denies that the theme was a unicorn but does not deny Josef saw something. Even though this analysis does make event variables available, in the case of (2.a), for example, the intend meaning does not seem to be that "Johnny did not see his father but he indeed saw someone/something".

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